

6(a)	energy transferred per (unit) charge (from electrical to other forms)	B1
6(b)	same/equal current (in X and Y)	B1
	$P = I^2 R$ (and $R_X > R_Y$) or $P = VI$ and $V_X > V_Y$	M1
	(so) X (dissipates more power)	A1
6(c)(i)	$I = Q / t$	C1
	$= 650 / 540$	A1
	$= 1.2 \text{ A}$	
6(c)(ii)	$V = W / Q$ or W / It	C1
	$= 4800 / 650$ or $4800 / (1.2 \times 540)$	A1
	$= 7.4 \text{ V}$	
	or	
	$V = P / I$ and $P = W / t$	(C1)
	$= 8.9 / 1.2$	(A1)
6(c)(iii)	$I = 4.5 + 1.2 (= 5.7 \text{ A})$	C1
	$9.0 = 7.4 + 5.7r$ or $9.0 = 5.7 (1.3 + r)$	A1
	$r = 0.28 \Omega$	